COMPONENT	FUNCTION	TECHNOLOGY USED	COMMUNICATION PROTOCOL	MAIN RESPONSIBILITY
VOICE API GATEWAY	Exposes the voice-service API to SAM Devices and forwards requests to the processing pipeline.	Node.js	REST over HTTPS; audio sent as binary or multipart payload	Device-facing access layer and request validation/routing.
VOICE AGENT	Orchestrates the complete voice interaction workflow.	Node.js	Internal REST/HTTP with JSON and audio payloads	Coordinates ASR, NLU, dialogue management, actions, and TTS.
AUTOMATIC SPEECH RECOGNITION	Converts recorded user speech into text.	Mozilla DeepSpeech, Node.js, Python	REST/HTTP; audio input and text/JSON output	Speech-to-text recognition using the trained ASR voice model.
TEXT-TO-SPEECH SERVICE	Converts the generated textual response into audio.	Mozilla TTS, Node.js, Python	REST/HTTP; text input and audio output	Speech synthesis using the configured TTS voice model.
NATURAL LANGUAGE UNDERSTANDING	Identifies intents and entities from the recognised text.	Rasa	REST/HTTP with JSON	Interprets the semantic meaning of the user request.
NATURAL LANGUAGE GENERATION	Determines the next dialogue step based on intent, entities, context, rules, and dialogue state. Also produces the textual response that will be returned to the user.	Rasa	REST/HTTP with JSON	Maintains conversation state and selects the appropriate response or action, then converts the selected dialogue response into user-facing text.
CUSTOM ACTIONS SERVICE	Executes domain-specific logic and obtains information from external services when needed.	Python, Rasa Actions	REST/HTTP or event-based interaction	Implements service-specific behaviours such as reminders, queries, or alerts.
CONTAINER ORCHESTRATION LAYER	Deploys, replicates, and scales service instances.	Kubernetes, Docker	Kubernetes internal networking and service discovery	Availability, scalability, isolation, and lifecycle management of microservices.